

Govind Kumar

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PROFESSIONAL SUMMARY

Results-driven **AI Engineer** pioneering edge-optimized Computer Vision pipelines for ATCC, TMCS, and Toll Automation. Proven expertise in orchestrating high-accuracy, multi-threaded inference at scale using PyTorch, TensorRT, and OpenVINO on resource-constrained hardware. Adept at transforming legacy architectures into robust, low-latency microservices to deliver scalable Intelligent Transportation Systems.

PROFESSIONAL EXPERIENCE

AI/ML Developer Feb 2026 – Present
Metro Infrasy Private Limited, Noida

- Architected a phased dual-model AI pipeline for ATCC, separating base vehicle localization from a centralized vehicle classification service to significantly enhance multi-camera system throughput.
- Engineered real-time Video Incident Detection Systems (VIDS) via **OpenVINO** for Edge AI; mapped 'stalled' vehicle detections to precise 'stop' event triggers to ensure highly reliable alerting logic.
- Guaranteed 99.9% stream uptime across complex deployments by implementing multi-threaded RTSP ingestion paired with automated network probing and robust reconnection logic.
- Developed configurable UDP-based radar monitoring applications, integrating strict software-level logic filters to dynamically suppress hardware-induced data anomalies and ensure pristine data ingestion.

Software Developer Trainee July 2025 – Jan 2026
Metro Infrasy Private Limited, Noida

- Collaborated in engineering high-accuracy **ANPR** and Automatic Vehicle Classification (AVC) systems for toll environments, structuring raw image capture pipelines to continuously feed training data for active model improvement.
- Spearheaded a major architectural redesign and complete UI/UX modernization, transforming monolithic code-bases into decoupled modular microservices to accelerate deployment cycles.
- Achieved **90.8% classification accuracy at 25 FPS** for complex YOLO-based models by executing rigorous quantization and pruning strategies.

Database Engineering Intern Aug 2023
EPlanet Technologies, New Delhi

- Optimized **MySQL** queries through strategic indexing and schema refinement, reducing execution time by **22%** for high-traffic datasets and improving overall pipeline latency.

KEY PROJECTS

Resource-Constrained Mobile Object Detection | *TensorFlow Lite, Android*

- Mitigated edge deployment bottlenecks by deploying a quantized SSD MobileNet architecture, achieving **85% accuracy at 30 FPS** for real-time mobile inference.

Low-Latency AI Attendance System | *OpenCV, Flask*

- Engineered a high-speed facial recognition system with **90% accuracy**; reduced manual tracking overhead by **70%** via optimized processing.

PUBLICATION

Mobile Real-Time Object Detection with SSD MobileNet 2025
International Journal of Scientific Research in Engineering and Management (IJSREM)
Authored peer-reviewed research analyzing resource-constrained deployment bottlenecks, focusing on optimizing SSD MobileNet architectures for high-speed edge inference.

EDUCATION

B.Tech | AI & Data Science
University School of Automation & Robotics
GGSIPU Delhi 2021–2025
GPA: 8.25/10

CERTIFICATIONS

- **Full Stack Developer** (GFG)
- **Data Analytics & Viz** (Forage)
- **Power BI for Beginners** (Skillup)